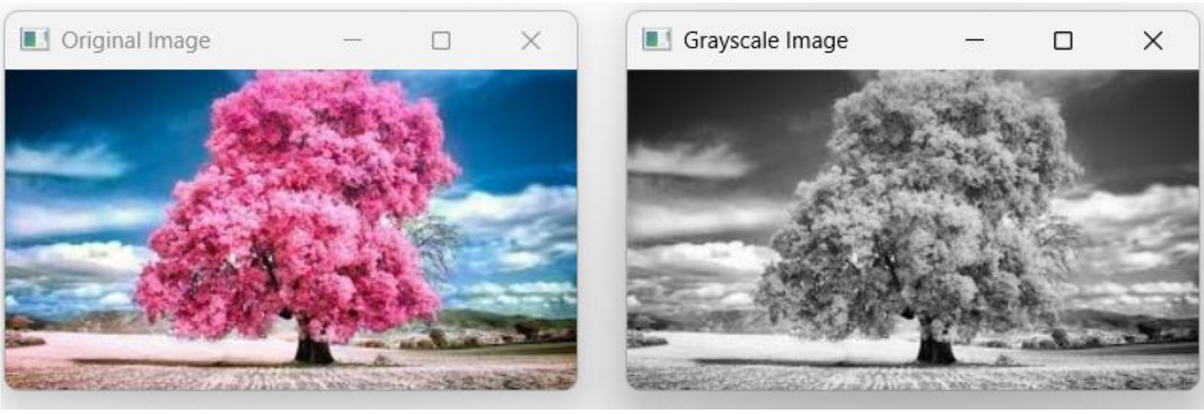
EXP 1:

EXP 1.py - C:/Users/hitha/OneDrive/Documents/CV EXP/EXP 1.py (3.14.3)

File Edit Format Run Options Window Help

```python
import cv2

# Read the image
img = cv2.imread(r"C:\Users\hitha\OneDrive\Pictures\Pictures\Picture CV.png")

# Check whether image is loaded
if img is None:
    print("Image not found!")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Display images
    cv2.imshow("Original Image", img)
    cv2.imshow("Grayscale Image", gray)

    # Save grayscale image
    cv2.imwrite("grayscale.jpg", gray)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

OUTPUT:



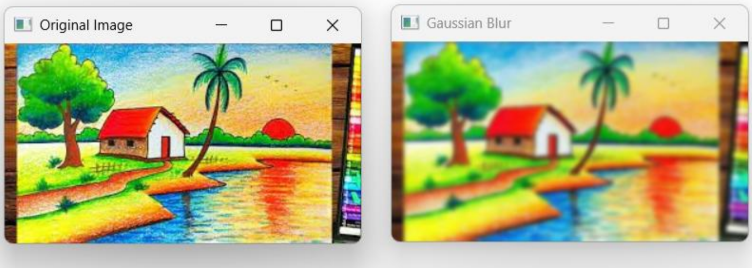
EXP 2:

File Edit Format Run Options Window Help

```python
import cv2

# Read image
img = cv2.imread(r"C:\Users\hitha\OneDrive\Pictures\Pictures\Picture2 Cv.jpg")

if img is None:
    print("Image not found!")
else:
    # Gaussian Blur
    blur = cv2.GaussianBlur(img, (9,9), 0)

    cv2.imshow("Original Image", img)
    cv2.imshow("Gaussian Blur", blur)

    cv2.imwrite("blur.jpg", blur)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

OUTPUT:

```
================================================================ RESTART: C:/Users/hitha/OneDrive/Documents/CV EXP/Exp 2.py ==
```



EXP 3:

File Edit Format Run Options Window Help

```python
import cv2

# Read image
img = cv2.imread(r"C:\Users\hitha\OneDrive\Pictures\Pictures\Picture3 CV.jpg")

if img is None:
    print("Image not found!")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Edge Detection
    edges = cv2.Canny(gray, 100, 200)

    cv2.imshow("Original Image", img)
    cv2.imshow("Canny Edge Detection", edges)

    cv2.imwrite("canny.jpg", edges)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

OUTPUT:

```
====================== RESTART: C:/Users/hitha/OneDrive/Documents/CV EXP/EXP 3.py =========
```

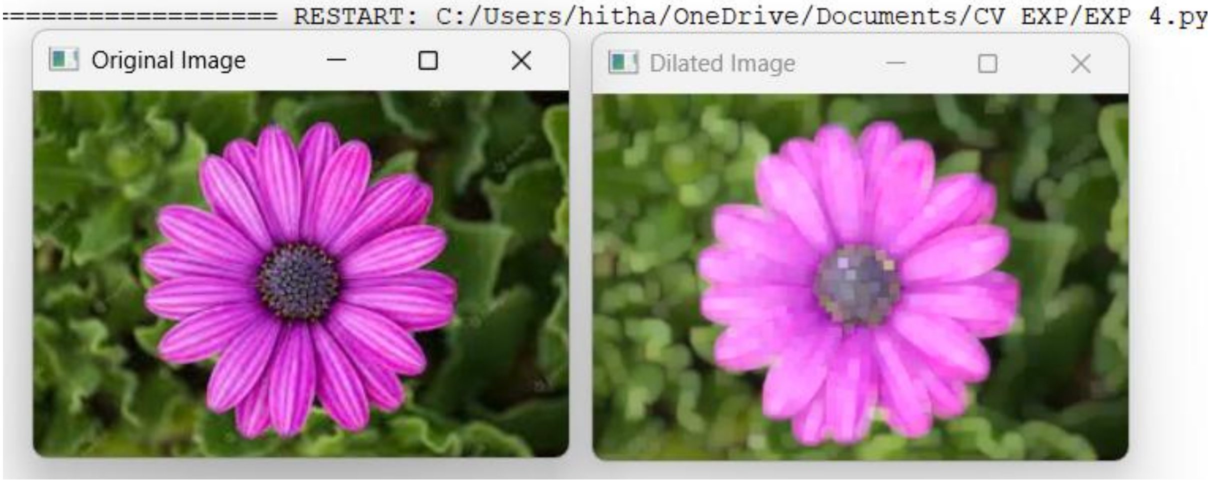
EXP 4

File Edit Format Run Options Window Help

```python
import cv2
import numpy as np

# Read image
img = cv2.imread(r"C:\Users\hitha\OneDrive\Pictures\Pictures\Picture4 CV.jpg")

if img is None:
    print("Image not found!")
else:
    # Create kernel
    kernel = np.ones((5,5), np.uint8)

    # Dilate image
    dilated = cv2.dilate(img, kernel, iterations=1)

    cv2.imshow("Original Image", img)
    cv2.imshow("Dilated Image", dilated)

    cv2.imwrite("dilated.jpg", dilated)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

OUTPUT:

```
================== RESTART: C:/Users/hitha/OneDrive/Documents/CV EXP/EXP 4.py
```

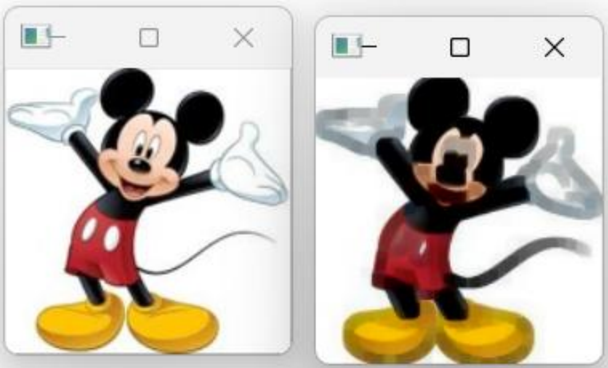


EXP 5:

File Edit Format Run Options Window Help

```
import cv2
import numpy as np

# Read image
img = cv2.imread(r"C:\Users\hitha\OneDrive\Pictures\Pictures\Picture5 CV.jpg")

if img is None:
    print("Image not found!")
else:
    # Create kernel
    kernel = np.ones((5,5), np.uint8)

    # Erode image
    eroded = cv2.erode(img, kernel, iterations=1)

    cv2.imshow("Original Image", img)
    cv2.imshow("Eroded Image", eroded)

    cv2.imwrite("eroded.jpg", eroded)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

OUTPUT:



EXP 8

File Edit Format Run Options Window Help

```
import cv2

# Read image
img = cv2.imread(r"C:\Users\hitha\OneDrive\Pictures\Pictures\Picture CV.png")

if img is None:
    print("Image not found!")
else:
    # Bigger image (2x)
    bigger = cv2.resize(img, None, fx=2, fy=2)

    # Smaller image (0.5x)
    smaller = cv2.resize(img, None, fx=0.5, fy=0.5)

    cv2.imshow("Original", img)
    cv2.imshow("Bigger Image", bigger)
    cv2.imshow("Smaller Image", smaller)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

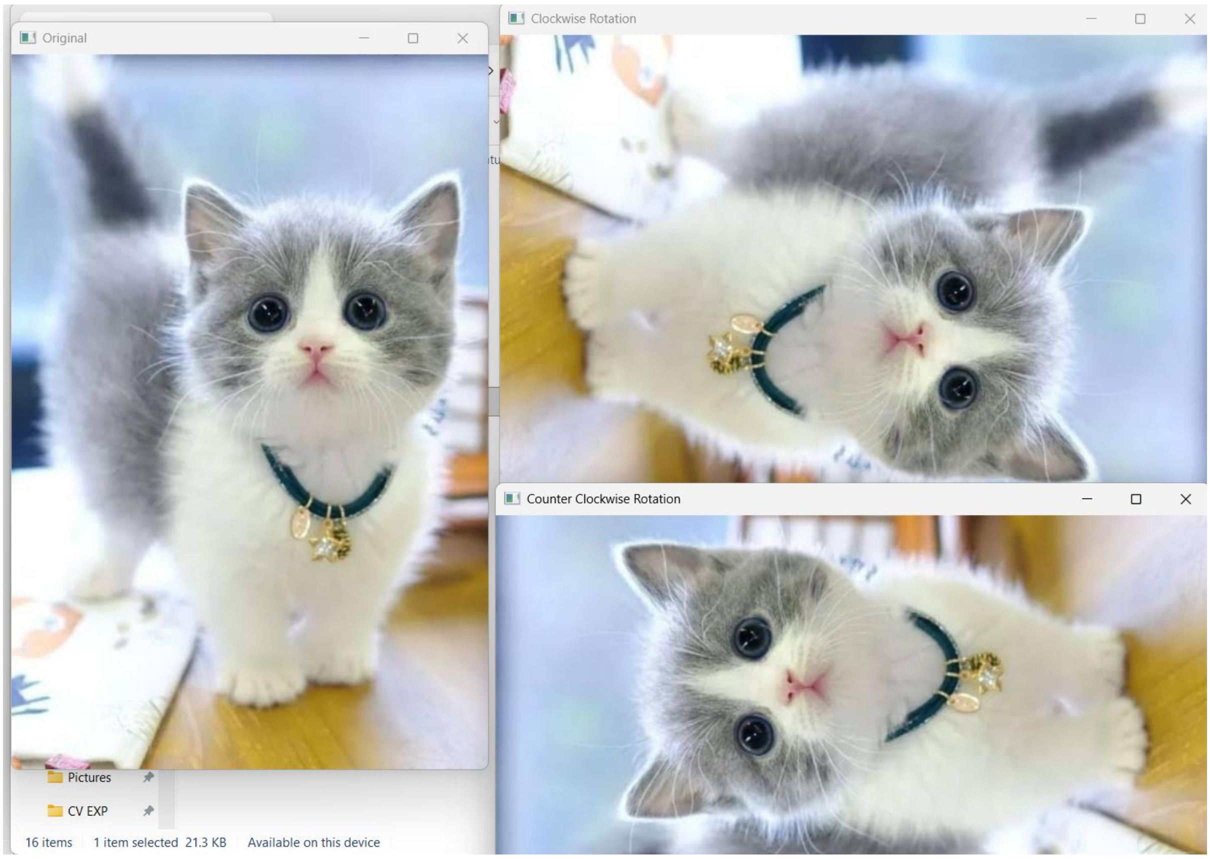
OUTPUT:

EXP 9

File Edit Format Run Options Window Help

```python
import cv2

# Read image
img = cv2.imread(r"C:\Users\hitha\OneDrive\Pictures\Pictures\Picture 9.jpg")

if img is None:
    print("Image not found!")
else:
    # Clockwise Rotation (90°)
    clockwise = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)

    # Counter Clockwise Rotation (90°)
    counter = cv2.rotate(img, cv2.ROTATE_90_COUNTERCLOCKWISE)

    cv2.imshow("Original", img)
    cv2.imshow("Clockwise Rotation", clockwise)
    cv2.imshow("Counter Clockwise Rotation", counter)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

OUTPUT:



EXP 10

File Edit Format Run Options Window Help

```python
import cv2
import numpy as np

# Read image
img = cv2.imread(r"C:\Users\hitha\OneDrive\Pictures\Pictures\Picture3 CV.jpg")

if img is None:
    print("Image not found!")
else:
    rows, cols = img.shape[:2]

    # Translation Matrix
    tx = 100    # Move Right
    ty = 50     # Move Down

    M = np.float32([[1, 0, tx],
                    [0, 1, ty]])

    moved = cv2.warpAffine(img, M, (cols, rows))

    cv2.imshow("Original Image", img)
    cv2.imshow("Translated Image", moved)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

OUTPUT:

```
========== RESTART: C:/Users/hitha/OneDrive/Documents/CV EXP/EXP 10.py =======
```